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IMAGING INFORMATICS AND ARTIFICIAL INTELLIGENCE

# Predicting post-lung transplant survival in systemic sclerosis using CT-derived features from preoperative chest CT scans

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## Abstract

**Objectives** The current understanding of survival prediction of lung transplant (LTx) patients with systemic sclerosis (SSc) is limited. This study aims to identify novel image features from preoperative chest CT scans associated with post-LTx survival in SSc patients and integrate them into comprehensive prediction models.

**Materials and methods** We conducted a retrospective study based on a cohort of SSc patients with demographic information, clinical data, and preoperative chest CT scans who underwent LTx between 2004 and 2020. This cohort consists of 102 patients (mean age, 50 years ± 10, 61% (62/102) females). Five CT-derived body composition features (bone, skeletal muscle, visceral, subcutaneous, and intramuscular adipose tissues) and three CT-derived cardiopulmonary features (heart, arteries, and veins) were automatically computed using 3-D convolutional neural networks. Cox regression was used to identify post-LTx survival factors, generate composite prediction models, and stratify patients based on mortality risk. Model performance was assessed using the area under the receiver operating characteristics curve (ROC-AUC).

**Results** Muscle mass ratio, bone density, artery–vein volume ratio, muscle volume, and heart volume ratio computed from CT images were significantly associated with post-LTx survival. Models using only CT-derived features outperformed all state-of-the-art clinical models in predicting post-LTx survival. The addition of CT-derived features improved the performance of traditional models at 1-year, 3-year, and 5-year survival prediction with maximum AUC scores of 0.77 (0.67–0.86), 0.85 (0.77–0.93), and 0.90 (95% CI: 0.83–0.97), respectively.

**Conclusion** The integration of CT-derived features with demographic and clinical features can significantly improve t post-LTx survival prediction and identify high-risk SSc patients.

## Key Points

***Question*** *What CT features can predict post-lung-transplant survival for SSc patients?*
***Finding*** *CT body composition features such as muscle mass, bone density, and cardiopulmonary volumes significantly predict survival.*
***Clinical relevance*** *Our individualized risk assessment tool can better guide clinicians in choosing and managing patients requiring lung transplant for systemic sclerosis.*

**Keywords** Body composition, Deep learning, Lung transplantation, Scleroderma, Survival analysis

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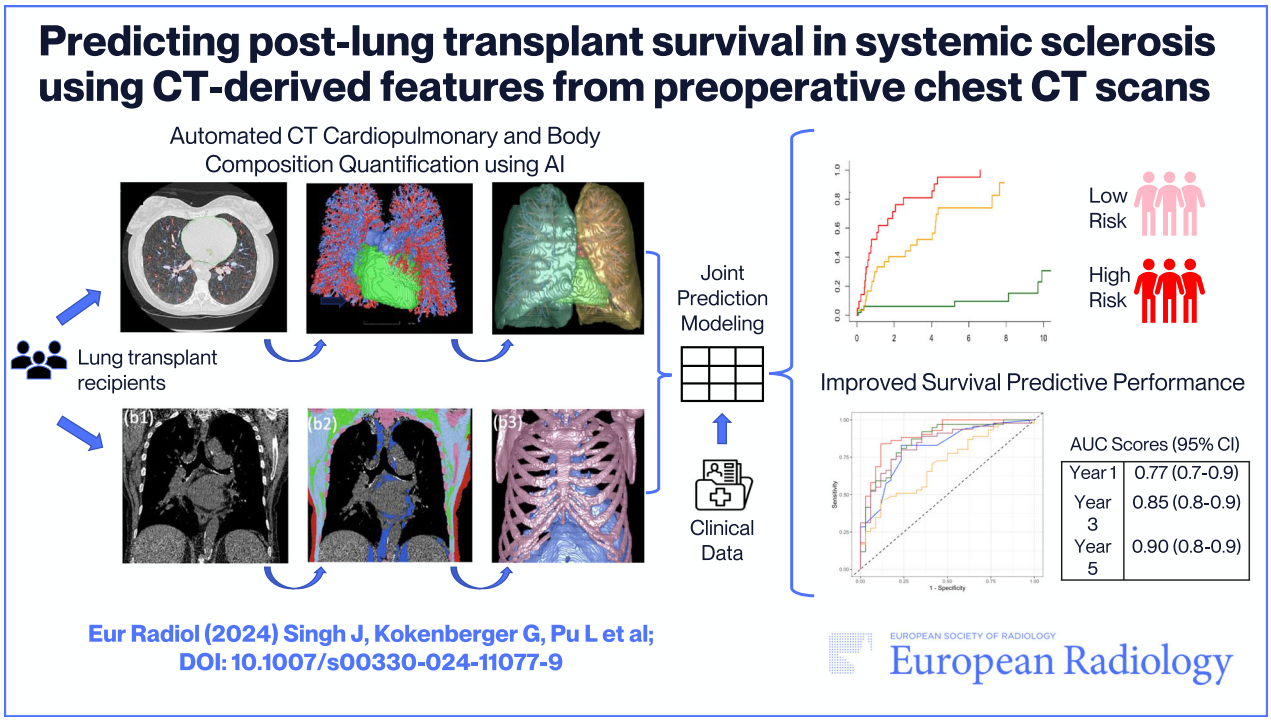
**Graphical Abstract**




## Introduction

Systemic sclerosis (SSc) is a connective tissue disorder characterized by extensive fibrosis of the skin and various visceral organs [1]. The pulmonary system is the second most common site of SSc manifestation, and lung complications are the leading cause of mortality in SSc patients [2, 3]. In situations where severe pulmonary destruction occurs, lung transplantation (LTx) may be indicated after ineffective therapy. LTx in patients with SSc only accounts for 0.9–1.2% of all LTx [2]. This low percentage can be attributed to the complex, multiorgan nature of the disease, which limits the number of centers performing LTx in SSc patients. Consequently, there is a pressing need for additional research to understand post-LTx survival factors and identify those who are suitable for this procedure [2–4].

Post-LTx survival is affected by a variety of factors, such as patient and donor characteristics, transplant operative features, and post-operative management decisions [5]. Pradere et al [3] identified female gender and pulmonary arterial hypertension (PAH) as significant risk factors for post-LTx survival in SSc patients. Oztekin et al [6] combined Cox proportional-hazards (CPH) models with a multi-layer perceptron to achieve survival prediction accuracies between 79% and 86% based on ten-fold cross-validation. Recently, machine learning (ML) techniques have been gaining special attention in LTx research for their superior predictive performance [7, 8].

Chest CT examinations are routinely performed pre-LTx to assess disease pathology and monitor patient health status. Undoubtedly, the CT images contain valuable information that can provide insights into a patient's overall health and disease pathology, which may be related to post-LTx complications and survival. Iyer et al [9] used CT-derived body composition features to predict esophageal cancer survival, achieving a 5-year prediction AUC of 0.861. In a meta-analysis on CT body composition, Elhakim et al [10] highlighted numerous successful clinical applications across various conditions, including multiple types of cancer, liver disease, cardiovascular disease, COVID-19, and more. However, few studies have utilized preoperative CT-derived features in LTx survival prediction. Horie et al [11] focused on non-SSc patients experiencing a decline in FEV1 and discovered that quantitative density metrics obtained from post-LTx CT scans were significant predictors (AUC = 0.74) of chronic lung allograft dysfunction (CLAD). In a separate study, Barbosa et al [12] found that models incorporating software-generated CT metrics predicted CLAD with higher accuracy (AUC = 0.817) than models based on

**Table 1** Univariate analysis of subject demographics for post-LTx survival in SSc patients ($n = 102$)

| Characteristic | Statistic | HR, (95% CI) | *p*-value |
|---|---|---|---|
| Age | | | |
| Mean (SD) | 51 (10) | 1.31 (0.98, 1.75) | 0.0656 |
| Range | 28–70 | | |
| Gender | | | |
| Female | 62/102 (60.8%) | 1.25 (0.71, 2.23) | 0.4408 |
| Male | 40/102 (39.2%) | | |
| BMI (kg/m$^2$) | | | |
| Mean (SD) | 24.9 (4.8) | 0.67 (0.50, 0.91) | 0.0116 |
| Range | 14.9–36.6 | | |
| Underweight (< 18.5) | 7/102 | 1.94 (0.83. 4.57) | 0.1271 |
| Normal (18.5–24.9) | 47/102 | 1.36 (0.78, 2.37) | 0.2826 |
| Overweight (25.0–29.9) | 30/102 | 0.75 (0.41, 1.40) | 0.3683 |
| Obese (> 29.9) | 18/102 | 0.49 (0.18, 1.37) | 0.1742 |
| Race | | | |
| White | 78/102 (76.5%) | 2.36 (1.00, 5.54) | 0.0489 |
| African American | 20/102 (19.6%) | 0.31 (0.11, 0.85) | 0.0234 |
| Other | 4/102 (3.9%) | – | – |
| Smoking status | | | |
| Former | 42/102 (41.2%) | 1.17 (0.67, 2.05) | 0.5943 |
| Never | 60/102 (58.8%) | | |
| Other characteristics | | | |
| Days between CT and transplant | | | |
| Mean (SD) | 119 (96) | | |
| Range | 2–366 | | |

Hazard ratios (HRs) of continuous variables are measured per unit
*SSc* systemic sclerosis, *SD* standard deviation

radiologist scoring and pulmonary function testing (AUC = 0.771) in a cohort of non-SSc LTx patients.

We investigated pre-LTx CT image features related to post-LTx survival in a cohort of 102 SSc patients. Deep learning algorithms were used to automatically segment and compute CT image features related to body composition and the cardiopulmonary system. Our objective was to identify novel image features associated with post-LTx survival in SSc patients and integrate them with demographic and clinical factors as a computer model to predict post-LTx survival. This model can also be used to stratify patients based on the risk of mortality in SSc patients and ultimately facilitate clinical decision-making in treating SSc patients.

## Materials and methods

### Study cohort

This retrospective study was approved by the University of Pittsburgh Institutional Review Board (IRB#: 20100067). We identified a cohort of 102 SSc patients who underwent LTx at the University of Pittsburgh Medical Center from 2004 to 2020. All patients met the inclusion criteria of SSc diagnosis, availability of chest CT imaging prior to LTx, and LTx due to SSc complications. The study cutoff date was January 28, 2021, with an average latest follow-up of 1475 days, ranging from 4 to 5328 days) post-LTx. Among these patients, 49.0% (50/102) died during the study period. Subject demographics had a reasonable representation of gender, race, and age (Table 1).

Pre-LTx non-contrast chest CT scans from the patients were performed on various multi-slice [13] CT scanners (Optima CT660 [GE Healthcare], Lightspeed 16 [GE Healthcare], Lightspeed Plus [GE Healthcare], and Lightspeed VCT [GE Healthcare]). Patients were positioned feet-first supine. Scan parameters include tube voltage 120 kV and tube current 302 ± 160 mAs. The CT images were reconstructed with a lung kernel as a matrix of 512 × 512 with a slice thickness ranging from 0.625 mm to 4 mm. The series with the maximum number of CT image slices were selected for the analyses.

### CT image features

Three-dimensional (3-D) convolutional neural networks (CNN) algorithms [14–17] were utilized to automatically segment and quantify five different body tissues and cardiopulmonary features depicted on CT images. These algorithms were validated in multiple studies and performed robustly on various slice thicknesses and CT scanners [13, 14, 17]. The classic Unet model, equipped with a grid-based 3-D patch sampling operation [17], was utilized to segment five different tissues on CT images. The five body tissues include visceral adipose tissue (VAT), subcutaneous adipose tissue (SAT), intramuscular adipose tissue (IMAT), skeletal muscle (SM), and bone (Fig. 1). For each segmented tissue, the volume (liters), mass (kg), and density were computed as previously described [9]. The average Hounsfield (HU) value was a surrogate for density and mass was estimated from HU values of segmented tissues using a validated equation [18]. To avoid collinearity, we use the ratio between mass and total mass in the analysis in place of mass. For example, the SM ratio is SM mass (kg) divided by the total mass of the body tissues in the chest (kg).

The cardiopulmonary structures included pulmonary arteries, veins, and the heart. Deep learning and computational differential geometry technologies were combined to identify arteries and veins. The utilization of the computational differential geometry method allows for the identification of small vessels. Our experiments on an independent test set ($n = 15$) showed no branch labeled by the radiologists was missed by the computer algorithm, and 98% of the intra-pulmonary vessel branches were correctly labeled [14]. Lung volumes were used to divide arteries and veins into separate categories of intrapulmonary and extrapulmonary arteries and veins,

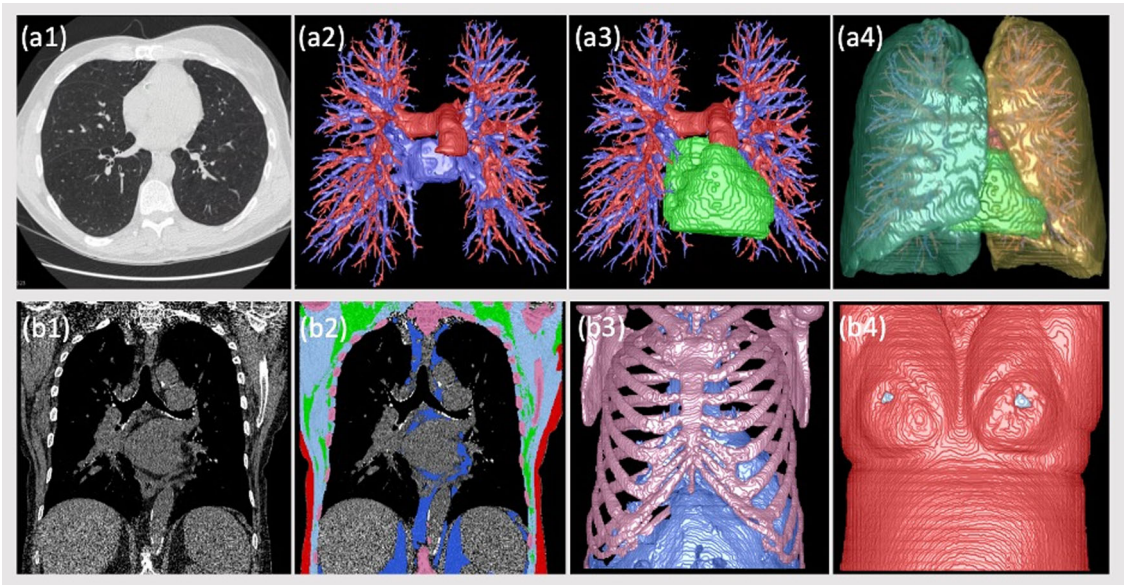



**Fig. 1** Our automated algorithm's segmentation results for various cardiopulmonary structures (top row) and body tissues (bottom row) depicted on non-contrast chest CT scans. Panel **a1** represents the axial view of a subject's chest CT scan, demonstrating the automatic segmentation of pulmonary vasculature (**a2**), heart (**a3**), and lungs (**a4**). Panel **b1** represents the coronal view of a subject's chest CT scan, demonstrating the automatic segmentation of body tissue in **b2**–**b4**. Five different body tissues include SAT (red), VAT (dark blue), IMAT (light green), SM (blue-gray), and bone (pink). Pulmonary arteries are segmented as red, the pulmonary veins are segmented as blue, and the heart is segmented as green

and their volumes were computed. Cardiopulmonary variables included pulmonary artery volume, pulmonary vein volume, and heart volume, which were automatically computed from the CT images (Fig. 1). We also computed volume ratios among the anatomical structures. For example, the heart ratio is the heart volume (L) divided by the total volume of the chest cavity calculated by the CNN algorithms.

### Clinical features

Subject characteristics and clinical features pertaining to preoperative variables were obtained from the medical record review. Subject demographics included age, gender, BMI, race, and smoking status (Table 1). In total, 64 clinical features were considered before preprocessing. Clinical features with missing values, lack of variance, or unreadable data were excluded from the analyses. After these preprocessing steps, 19 clinical features remained (Table 2).

### Statistical analysis

Survival analysis was conducted using the CPH model. All categorical features were one-hot encoded, and all linear features were *z*-score standardized. For patients who survived past the cutoff date, right censoring was applied. Initially, all variables were evaluated in a univariate analysis. HRs of continuous variables were measured per unit. Before multivariate analysis, features with a variable inflation factor greater than five were removed to avoid high collinearity between features. Multivariate CPH models were then constructed using the backward-stepwise regression approach, in which covariates are eliminated iteratively based on their association strength until all remaining covariates achieve statistical significance. Time-point survival classification was performed on the cohort at one, three, and five years post-LTx. The proportional hazards assumption was verified at each step.

To ensure robustness, time-point analysis was validated using 5-fold cross-validation. The performance of the models was evaluated using the concordance index (*c*-index), Brier score, and area under the receiver operating characteristics curve (ROC-AUC). These models used only pre-LTx features, excluding variables like hospital length of stay or surgical features, to simulate pre-LTx screening ability. Specialized non-parametric tests designed for time-dependent ROC-AUCs [19] were used to compare model performances because DeLong's test is less powerful in censored time-to-event analysis and DeLong's assumption could be violated by our data [20, 21]. For univariate analyses, a significance level of 0.05 was considered statistically significant. For model ROC-AUC comparison with 5-fold cross-validation, Bonferroni's correction for multiple comparison testing (five tests) was applied, and a *p*-value less than 0.01 was considered statistically significant [22]. Data preprocessing was conducted using Python 3.9 while statistical analysis was performed in R3.4.1.

**Table 2** Univariate analysis of clinical features for post-LTx survival in SSc patients

| Characteristic | HR | CI, (95%) | Deceased, ($n = 50$) | Alive, ($n = 52$) | *p*-value |
|---|---|---|---|---|---|
| Age-adjusted CCI | 1.48 | (1.14, 1.91) | 3.7 ± 1.2 | 2.9 ± 1.0 | 0.0029 |
| Tracheostomy | 2.13 | (1.20, 3.77) | 25 (50%) | 18 (35%) | 0.0097 |
| Calcinosis | 1.79 | (0.98, 3.27) | 15 (30%) | 6 (12%) | 0.0602 |
| Pulmonary arterial HTN | 0.42 | (0.17, 1.09) | 45 (90%) | 50 (96%) | 0.0775 |
| Atrial fibrillation | 1.66 | (0.90, 3.06) | 15 (30%) | 14 (27%) | 0.1029 |
| mPAP (mmHg) | 0.81 | (0.61, 1.08) | 34.1 ± 10.7 | 36.8 ± 11.7 | 0.1541 |
| Preop FEV1/FVC | 0.83 | (0.62, 1.12) | 0.8 ± 0.1 | 0.8 ± 0.2 | 0.2194 |
| SSc class—systemic | 1.43 | (0.75, 2.75) | 12 (24%) | 6 (12%) | 0.2779 |
| Articular involvement | 1.35 | (0.77, 2.38) | 20 (40%) | 16 (31%) | 0.3004 |
| Lung allocation score | 0.88 | (0.65, 1.18) | 57.0 ± 19.2 | 61.8 ± 18.4 | 0.3965 |
| Previous chest procedure | 0.78 | (0.38, 1.60) | 9 (18%) | 13 (25%) | 0.4961 |
| Telangiectasia | 1.21 | (0.69, 2.11) | 28 (56%) | 24 (46%) | 0.5071 |
| SSc class—limited systemic | 0.83 | (0.47, 1.45) | 23 (46%) | 27 (52%) | 0.5097 |
| Esophageal dysmotility | 1.33 | (0.41, 4.29) | 47 (94%) | 48 (92%) | 0.6309 |
| Sclerodactyly | 1.16 | (0.64, 2.10) | 34 (68%) | 36 (69%) | 0.6333 |
| SSc class—sine | 0.79 | (0.28, 2.20) | 4 (8%) | 7 (13%) | 0.6495 |
| PAWP (mmHg) | 1.02 | (0.78, 1.35) | 10.6 ± 6.5 | 10.5 ± 4.5 | 0.8646 |
| SSc class—diffuse cutaneous | 1.04 | (0.53, 2.03) | 11 (22%) | 12 (23%) | 0.9101 |
| Interstitial lung disease | 1.00 | (0.56, 1.79) | 31 (62%) | 38 (73%) | 0.9946 |

HRs of continuous variables are measured per unit
*SSc* systemic sclerosis, *CCI* Charlson comorbidity index, *HTN* hypertension, *mPAP* mean pulmonary arterial pressure, *FEV1/FVC* forced expiratory volume/forced vital capacity, *PAWP* pulmonary arterial wedge pressure

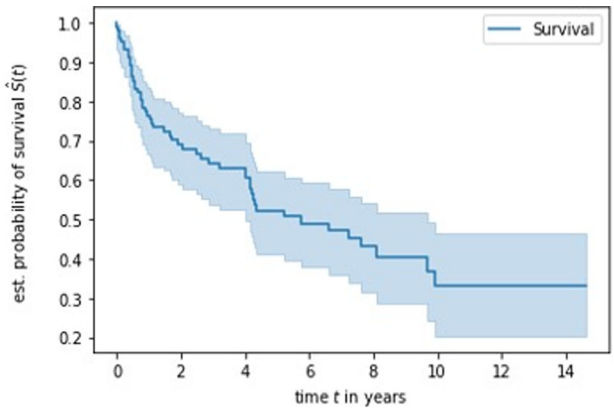



**Fig. 2** Kaplan–Meier survival curve of the study cohort ($n = 102$). Median survival for the cohort is 6.6 years post-transplant

### Risk stratification

The study cohort was stratified into low, intermediate, and high-risk groups based on mortality risk scores generated by the best-performing multivariate CPH model. We opted to identify risk scores for year-5 survival using the best model specifically; however, similar results can be generated at any time point with the same methods. The low-risk group included patients in the bottom 50th percentile of risk scores at year 5. The high-risk group included patients in the top 20th percentile. The remaining patients formed the intermediate-risk group. The log-rank test was used to evaluate inter-group differences in survival distributions and cumulative incidence function graphs were utilized to visualize the risk groups. Finally, Kaplan–Meier survival curves were generated to illustrate the feature values for each risk group.

## Results

### Kaplan–Meier survival analysis

Fifty patients (49.0%) died before the study cutoff date of January 28, 2021. The 1-year, 3-year, and 5-year post-LTx survival rates were 76.3%, 64.3%, and 52.2%, respectively, based on the Kaplan–Meier survival analysis (Fig. 2). The median survival of the cohort was 6.6 years post-LTx.

### Subject demographics

Among the demographic factors, the most significant predictor of survival was the subject's BMI (kg/m$^2$) with an HR of 0.67 (95% confidence interval (CI): 0.50–0.91, $p = 0.0116$) indicating a higher BMI was protective in terms of mortality (Table 1). Race was also a significant independent predictor of survival. African American patients demonstrated a protective HR of 0.31 (95% CI: 0.11–0.85, $p = 0.0489$), while white patients demonstrated a hazardous HR of 2.36 (95% CI: 1.00–5.54, $p = 0.0234$). In the survival composite model incorporating all demographic variables, BMI (HR = 0.66, 95% CI: 0.47–0.91,

**Table 3** Univariate analysis of CT-derived features for post-LTx survival in SSc patients

| Characteristic | HR | CI, (95%) | Deceased | Alive | *p*-value |
|---|---|---|---|---|---|
| Cardiopulmonary features | | | | | |
| Extrapulmonary artery vol | 1.04 | (0.78, 1.38) | 0.08 ± 0.04 | 0.08 ± 0.02 | 0.8025 |
| Extrapulmonary vein vol. | 0.91 | (0.7, 1.19) | 0.12 ± 0.06 | 0.13 ± 0.04 | 0.5004 |
| Intrapulmonary artery vol. | 1.16 | (0.88, 1.54) | 0.08 ± 0.04 | 0.08 ± 0.03 | 0.2984 |
| Intrapulmonary vein vol. | 0.83 | (0.63, 1.10) | 0.12 ± 0.06 | 0.14 ± 0.05 | 0.2007 |
| Artery–vein vol. ratio | 1.48 | (1.21, 1.96) | 0.73 ± 0.28 | 0.55 ± 0.19 | 0.0002 |
| Heart vol. | * | * | 0.69 ± 0.29 | 0.66 ± 0.21 | * |
| Heart ratio | 1.28 | (0.94, 1.74) | 0.08 ± 0.04 | 0.07 ± 0.02 | 0.1156 |
| Body composition features | | | | | |
| VAT volume | 1.18 | (0.9, 1.55) | 0.49 ± 0.51 | 0.42 ± 0.37 | 0.2202 |
| VAT density | 1.06 | (0.78, 1.43) | − 85.1 ± 11.5 | − 86.4 ± 8.6 | 0.7160 |
| VAT ratio | 1.21 | (0.92, 1.58) | 0.05 ± 0.04 | 0.04 ± 0.03 | 0.1717 |
| SAT volume | 1.26 | (0.94, 1.69) | 2.69 ± 1.73 | 2.52 ± 1.32 | 0.1148 |
| SAT density | 0.97 | (0.72, 1.3) | − 84.5 ± 19.2 | − 87.1 ± 17.2 | 0.8183 |
| SAT ratio | 1.27 | (0.95, 1.71) | 0.29 ± 0.12 | 0.27 ± 0.1 | 0.1096 |
| IMAT volume | 1.10 | (0.81, 1.49) | 0.33 ± 0.19 | 0.35 ± 0.2 | 0.5475 |
| IMAT density | 1.02 | (0.76, 1.35) | − 85.5 ± 12.6 | − 85.4 ± 8.3 | 0.9152 |
| IMAT ratio | 1.12 | (0.84, 1.51) | 0.04 ± 0.02 | 0.04 ± 0.02 | 0.4339 |
| SM volume | 0.80 | (0.62, 1.04) | 3.07 ± 0.99 | 3.31 ± 0.83 | 0.1008 |
| SM density | 0.74 | (0.54, 1.02) | 24.5 ± 13.5 | 26.4 ± 8.5 | 0.0669 |
| SM ratio | 0.71 | (0.52, 0.97) | 0.4 ± 0.09 | 0.43 ± 0.09 | 0.0332 |
| Bone volume | 0.87 | (0.68, 1.13) | 1.31 ± 0.37 | 1.34 ± 0.32 | 0.3095 |
| Bone density | 0.70 | (0.52, 0.96) | 295.5 ± 61.1 | 314.7 ± 53.8 | 0.0270 |
| Bone ratio | 0.87 | (0.65, 1.17) | 0.22 ± 0.06 | 0.22 ± 0.04 | 0.3493 |
| Total mass | 1.00 | (0.75, 1.34) | 9.12 ± 2.91 | 9.25 ± 2.25 | 0.9733 |

HRs of continuous variables are measured per unit
*SSc* systemic sclerosis, *VAT* visceral adipose tissue, *SAT* subcutaneous adipose tissue, *IMAT* intramuscular adipose tissue, *SM* skeletal muscle
* Features failed the non-proportional hazards test

$p = 0.0115$) and the African American Race (HR = 0.30, 95% CI: 0.11–0.84, $p = 0.0220$) remained statistically significant. The composite demographic model termed the Demographic model, achieved concordance indexes of 0.581, 0.631, and 0.640 for survival prediction at years 1, 3, and 5 post-LTx. The demographic model also achieved Brier scores of 0.175, 0.214, and 0.220 at years 1, 3, and 5 post-LTx. In the 5-fold cross-validation, the demographic model had a mean AUC of 0.61 (95% CI: 0.47–0.74) for predicting year-1 survival, 0.67 (95% CI: 0.55–0.79) for predicting year-3 survival, and 0.70 (95% CI: 0.59–0.82) for predicting year-5 survival.

### Clinical features

Age-adjusted CCI and tracheostomy were significantly associated with survival with HRs of 1.48 (95% CI: 1.14–1.91, $p = 0.0029$) and 2.13 (95% CI: 1.20–3.77, $p = 0.0097$), respectively, in the univariate analysis (Table 2). In the composite model incorporating all clinical features, age-adjusted CCI ($p < 0.05$) remained statistically significant. Notably, calcinosis (HR = 4.50, 95% CI: 1.23–6.73, $p = 0.0038$) and PAH (HR = 0.65, 95% CI: 0.43–0.92, $p = 0.0324$) emerged as significant predictors of survival in the composite model. The composite clinical model, termed the clinical model, achieved concordance indexes of 0.578, 0.621, and 0.647 in survival prediction at years 1, 3, and 5 post-LTx. The clinical model also achieved Brier scores of 0.172, 0.203, and 0.182 at years 1, 3, and 5 in survival prediction. In the 5-fold cross-validation, the clinical model had a mean AUC of 0.62 (95% CI: 0.50–0.75) for predicting year-1 survival, 0.71 (95% CI: 0.60–0.82) for predicting year-3 survival, and 0.81 (95% CI: 0.72–0.91) for predicting year-5 survival.

### CT-derived features

Among the ten cardiopulmonary parameters, the artery–vein ratio (HR = 1.48, 95% CI: 1.21–1.96, $p = 0.0002$) demonstrated predictive value for post-LTx survival (Table 3). Due to a violation of the proportional

**Table 4** Year-1, -3, and -5 post-LTx survival model performance for SSc patients ($n = 102$)

| Model | Brier score | *c*-index | AUC, (95% CI) | Variables included |
|---|---|---|---|---|
| Year 1 prediction scores | | | | |
| Demographic | 0.175 | 0.581 | 0.61 (0.47–0.74) | BMI and race |
| Clinical | 0.172 | 0.578 | 0.62 (0.50–0.75) | ACCI, trach, calcinosis, and PAH |
| Demo–clinical | 0.173 | 0.603 | 0.65 (0.53–0.77) | BMI, race, ACCI, trach, calcinosis, and PAH |
| CT-features | 0.159 | 0.737 | 0.77 (0.67–0.86) | Bone density, AV ratio, heart ratio, muscle volume, and muscle ratio |
| Aggregate | 0.163 | 0.694 | 0.75 (0.65–0.85) | BMI, race, ACCI, trach, calcinosis, PAH, bone density, AV ratio, heart ratio, muscle volume, and muscle ratio |
| Year 3 prediction scores | | | | |
| Demographic | 0.214 | 0.631 | 0.67 (0.55–0.79)[*,†] | BMI and race |
| Clinical | 0.203 | 0.621 | 0.71 (0.60–0.82) | ACCI, trach, calcinosis, and PAH |
| Demo–clinical | 0.191 | 0.661 | 0.75 (0.64–0.86) | BMI, race, ACCI, trach, calcinosis, and PAH |
| CT-features | 0.166 | 0.767 | 0.85 (0.77–0.93) | Bone density, AV ratio, heart ratio, muscle volume, and muscle ratio |
| Aggregate | 0.160 | 0.737 | 0.83 (0.75–0.92) | BMI, race, ACCI, trach, calcinosis, PAH, bone density, AV ratio, heart ratio, muscle volume, and muscle ratio |
| Year 5 prediction scores | | | | |
| Demographic | 0.220 | 0.640 | 0.70 (0.59–0.82)[*] | BMI and race |
| Clinical | 0.182 | 0.647 | 0.81 (0.72–0.91) | ACCI, trach, calcinosis, and PAH |
| Demo–clinical | 0.167 | 0.686 | 0.84 (0.75–0.93) | BMI, race, ACCI, trach, calcinosis, and PAH |
| CT-features | 0.163 | 0.767 | 0.86 (0.77–0.94) | Bone density, AV ratio, heart ratio, muscle volume, and muscle ratio |
| Aggregate | 0.134 | 0.752 | 0.90 (0.83–0.97) | BMI, race, ACCI, trach, calcinosis, PAH, bone density, AV ratio, heart ratio, muscle volume, and muscle ratio |

Maximum scores in each time-point are boldened
*SSc* systemic sclerosis, *ACCI* age-adjusted Charlson comorbidity index, *Trach* tracheostomy, *AV* artery–vein, *CPH* cox proportional hazards model
[*] *p*-value < 0.01 compared to the aggregate model
[†] *p*-value < 0.01 compared to CT-Features model

hazards assumption, heart volume was excluded from the analysis. In the composite model incorporating all cardiopulmonary features, heart ratio emerged as another significant predictor (HR = 1.53, 95% CI: 1.14–2.05, $p = 0.0051$).

Among the 16 body composition parameters examined, bone density (HR = 0.70, 95% CI: 0.52–0.96, $p = 0.0270$) and SM ratio (HR = 0.71, 95% CI: 0.52–0.97, $p = 0.0332$) demonstrated a strong protective relationship with post-LTx survival in the univariate analysis (Table 3). In the composite model incorporating all body composition features, muscle volume (HR = 0.76, 95% CI: 0.61–0.89, $p = 0.042$) emerged as an additional protective feature, while muscle ratio did not reach statistical significance. The composite CT-derived features model of cardiopulmonary and body compositional features, termed the CT-features model, achieved concordance indexes of 0.737, 0.767, and 0.767 for survival prediction at years 1, 3, and 5 post-LTx. The CT model also achieved Brier scores of 0.159, 0.166, and 0.163 at years 1, 3, and 5 post-LTx. In the 5-fold cross-validation, the CT model had a mean AUC of 0.77 (95% CI: 0.67–0.86) for predicting year-1 survival, 0.85 (95% CI: 0.77–0.93) for predicting year-3 survival, and 0.86 (95% CI: 0.77–0.94) for predicting year-5 survival.

### Integrative prediction model

Two composite models were developed, termed demo–clinical and aggregate models, to combine significant features from the above groups (Table 4). The demo–clinical model uses all significant pre-LTx demographic and clinical/surgical features to simulate traditional LTx survival models. The aggregate model incorporates all significant pre-LTx features including CT-derived features to achieve the maximum predictive performance and highlight the advantage of CT-derived features over traditional survival models.

The demo–clinical model achieved concordance indexes of 0.603, 0.661, and 0.686 for survival prediction at years 1, 3, and 5 post-LTx. The demo–clinical model also achieved Brier scores of 0.173, 0.191, and 0.167 at years 1, 3, and 5 post-LTx. In the 5-fold cross-validation, the demo–clinical model had a mean AUC of 0.65 (95% CI: 0.53–0.77) for predicting year-1 survival, 0.75 (95% CI: 0.64–0.86) for predicting year-3 survival, and 0.84 (95% CI: 0.75–0.93) for predicting year-5 survival (Table 4).

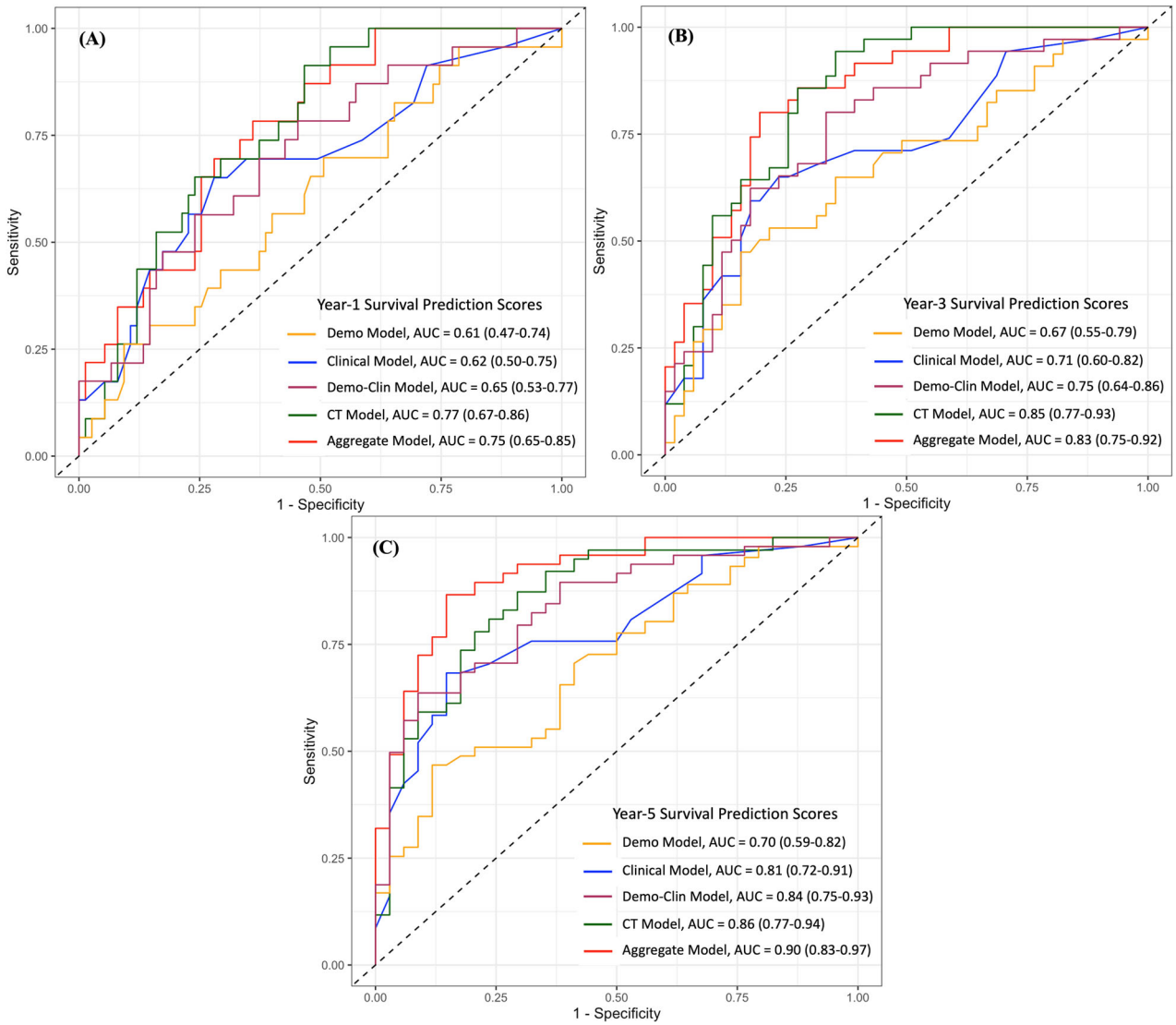



**Fig. 3** Three timepoint ROC curve plots at 1-year (**A**), 3-years (**B**), and 5-years (**C**) post-transplant comparing the models' ROC-AUC in 5-fold cross-validation. The Demo or demographic model includes BMI and Race. The Clinical model includes ACCI, tracheostomy, calcinosis, and PAH. The demo–clinical model includes the features included in the demo and clinical model. The CT model includes significant CT-derived features including bone density, intrapulmonary artery–vein volume ratio, heart volume ratio, muscle volume, and muscle mass ratio. The aggregate model combines all of these features listed into one model

The aggregate model achieved concordance indexes of 0.694, 0.737, and 0.752 for survival prediction at years 1, 3, and 5 post-LTx. The aggregate model also achieved Brier scores of 0.163, 0.160, and 0.134 at years 1, 3, and 5 post-LTx. In the 5-fold cross-validation, the aggregate model had a mean AUC of 0.75 (95% CI: 0.65–0.85) for predicting year-1 survival, 0.83 (95% CI: 0.75–0.92) for predicting year-3 survival, and 0.90 (95% CI: 0.83–0.97) for predicting year-5 survival (Table 4).

The CT-Features model outperforms the demographic, clinical, and demo–clinical models in 1-, 3-, and 5-year post-LTx survival prediction in brier scores, concordance index, and ROC-AUC (Fig. 3 and Table 4). Similarly, the aggregate model outperforms all other models in all metrics and achieves statistically significant improvements over demographic models in 3- ($p = 0.0067$) and 5-year ($p = 0.0032$) survival prediction ($p < 0.01$; Bonferroni adjusted), underscoring the advantage of CT-derived features in survival prediction (Fig. 3 and Table 4). Interestingly, the CT-Features model outperforms the aggregate model in year 1 and year 3 survival prediction, indicating that CT features alone can serve as predictors in survival models.

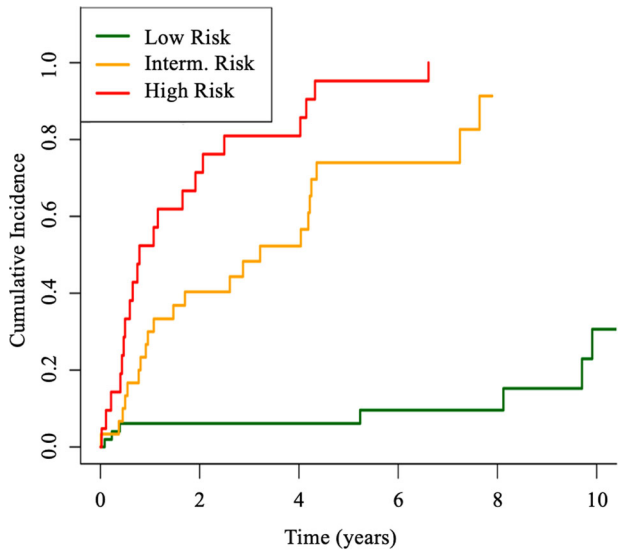



**Fig. 4** Cumulative incidence plot of post-LTx mortality stratified by patient risk stratum. Stratification was conducted by labeling the top 20th percentile as high-risk, the bottom 50th percentile as low-risk, and patients remaining as intermediate-risk

### Risk stratification

At 5 years post-LTx, 13.7%, 73.3%, and 100% of the patients identified as low ($n = 51$), intermediate ($n = 30$), and high-risk ($n = 21$) had died. The HR related to death 5 years post-LTx for the low, intermediate, and high-risk patients were 0.05 (95% CI: 0.02–0.14), 2.47 (95% CI: 1.39–4.42), and 6.61 (95% CI: 3.59–12.16), respectively (Fig. 4). Based on the log-rank analysis, the groups represent significantly different survival distributions ($p < 0.001$). Individual body composition (bone density, muscle volume, and muscle mass ratio), cardiopulmonary (artery–vein volume ratio), demographic (BMI), and clinical (tracheostomy and age-adjusted CCI) features were successfully stratified into low, intermediate, and high-risk (Figs. 5 and 6).

## Discussion

While LTx offers significant improvement in quality of life for patients suffering from SSc with pulmonary complications, LTx can have poor post-operative survival rates. Therefore, enhanced screening techniques and risk assessment tools may help identify those who are suitable for this procedure and facilitate rehabilitation based on the identified factors, ultimately improving survival and outcomes for SSc patients. We developed multiple models based on demographic, clinical, and radiographic features that successfully predicted mortality and identified patients at risk of mortality post-LTx. The inclusion of radiographic features improved the performance of the prediction models compared to traditional models based on demographic and clinical features.

Prior studies in this field mainly relied on patient demographic and clinical variables [3, 6–8], which yielded insightful results but had limitations related to data availability and prediction accuracy. While two studies used CT image features to predict post-LTx mortality and graft dysfunction [11, 12], they focused on metrics derived from post-LTx CT scans and limited their analysis to lung-specific CT metrics for post-LTx survival prediction. Importantly, these studies did not identify significant risks that could be assessed pre-LTx, eliminating any pre-LTx risk assessment potential. In contrast, our study took a broader approach by incorporating a detailed assessment of body composition tissues, such as fat volumes, muscle and bone densities, as well as cardiopulmonary characteristics of the heart, pulmonary arteries, and veins. In addition, to our knowledge, none of the CT-based predictive studies focused on SSc lung disease. Our objective was to capture the systemic nature of SSc by analyzing multiple organs and body composition before LTx. The comprehensive models we developed assess whole-body health to predict post-LTx survival successfully.

We found that certain radiographic features independently predicted post-LTx survival. Specifically, decreased bone density and muscle ratio emerged as predictors of poor post-LTx survival, indicating that a subject's frailty or sarcopenic status before LTx is predictive of survival. These negative prognostic factors have been validated in numerous body composition analysis applications, especially in demonstrating sarcopenia's association with increased mortality [10]. Additionally, the ratio between total intrapulmonary artery and vein volume was predictive of post-LTx survival, which may indicate the presence of cardiopulmonary abnormalities such as PAH that act as negative prognostic factors. We speculate that this ratio may reflect PAH, based on our results and a previous study that identified PAH as a risk factor for survival in SSc LTx [3]. In the multivariate analysis, an increased heart ratio also emerged as a significant risk factor. This CT feature may describe cardiomegaly, where the heart volume is enlarged relative to the total thoracic volume. This explanation would account for the hazardous risk finding. However, further studies are needed to clarify the exact pathophysiology behind the intrapulmonary artery–vein ratio and heart ratio.

Notably, the lung allocation score (LAS), a comprehensive survival estimate based on illness severity and recipient characteristics, was non-significant in univariate testing. In a recent review of LAS, Parker et al found that LAS is inaccurate, explaining only 56% of the variation in observed outcomes from more recent data [23]. Parker et al also note that LAS was based on two survival models

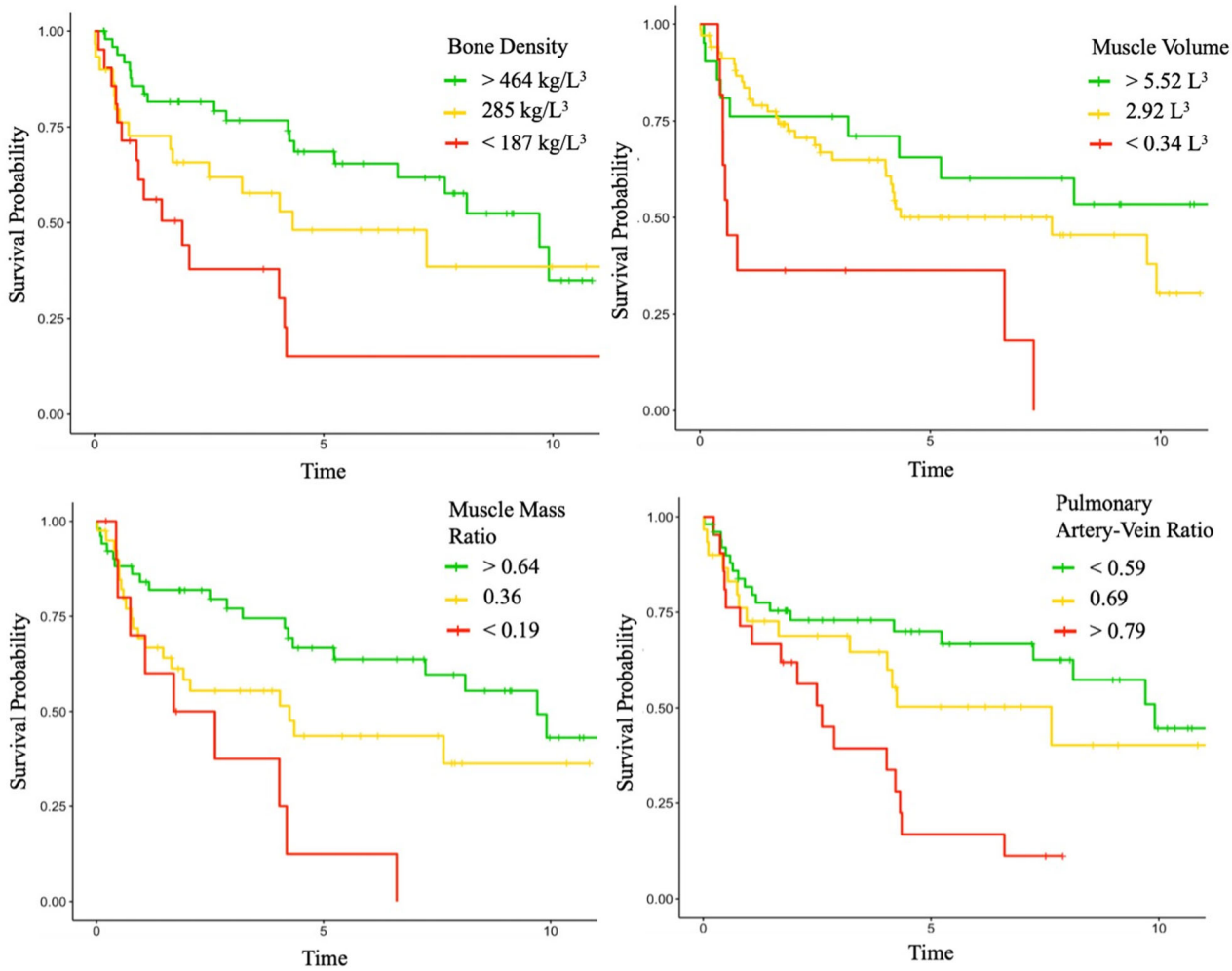



**Fig. 5** Kaplan–Meier curves of significant CT-Derived feature stratified by patient risk stratum. Threshold values are given that determine stratum cutoffs for the total cohort. Muscle mass ratio is the ratio between muscle mass (kg) and total mass (kg)

using data from 2005 to 2008 and has not been updated since. Considering these findings and the fact that LTx in SSc patients is a rare procedure that has been gaining traction in recent years, it is not surprising that the LAS score was not predictive of survival in our cohort.

Survival models using solely radiographic features (referred to as the CT-feature model) and models augmenting demographic and clinical features with radiographic features (referred to as the aggregate model) outperformed traditional models survival prediction 1 year, 3 years, and 5 years after transplantation. Our findings support the importance of including radiographic features as prognostic tools for SSc patient evaluation pre-LTx. CT-derived radiographic features may provide an objective assessment of a patient's body composition and general health or changes in health, which could inform an LTx risk assessment and ultimately improve LTx outcomes.

In contrast to previous studies [6–8, 12], our approach did not involve the use of complex ML algorithms like random survival forest or SVM. While these ML models may offer enhanced predictive capability, they often lack interpretability, which was a key objective of our study and a crucial aspect of clinical application. The interpretability issue arises due to the "black-box" nature of ML models, which are optimized for performance but lack transparency in identifying specific risk factors [24]. We generated both univariate and multivariate HRs using well-established time-to-event CPH regression models that allowed for a straightforward interpretation of the results. Moreover, the risk scores derived from our CPH models facilitated the stratification of patients into low, intermediate, and high-risk categories. Our focus on using CPH models enabled practical application in a clinical setting. In this context, having specific risk ranges

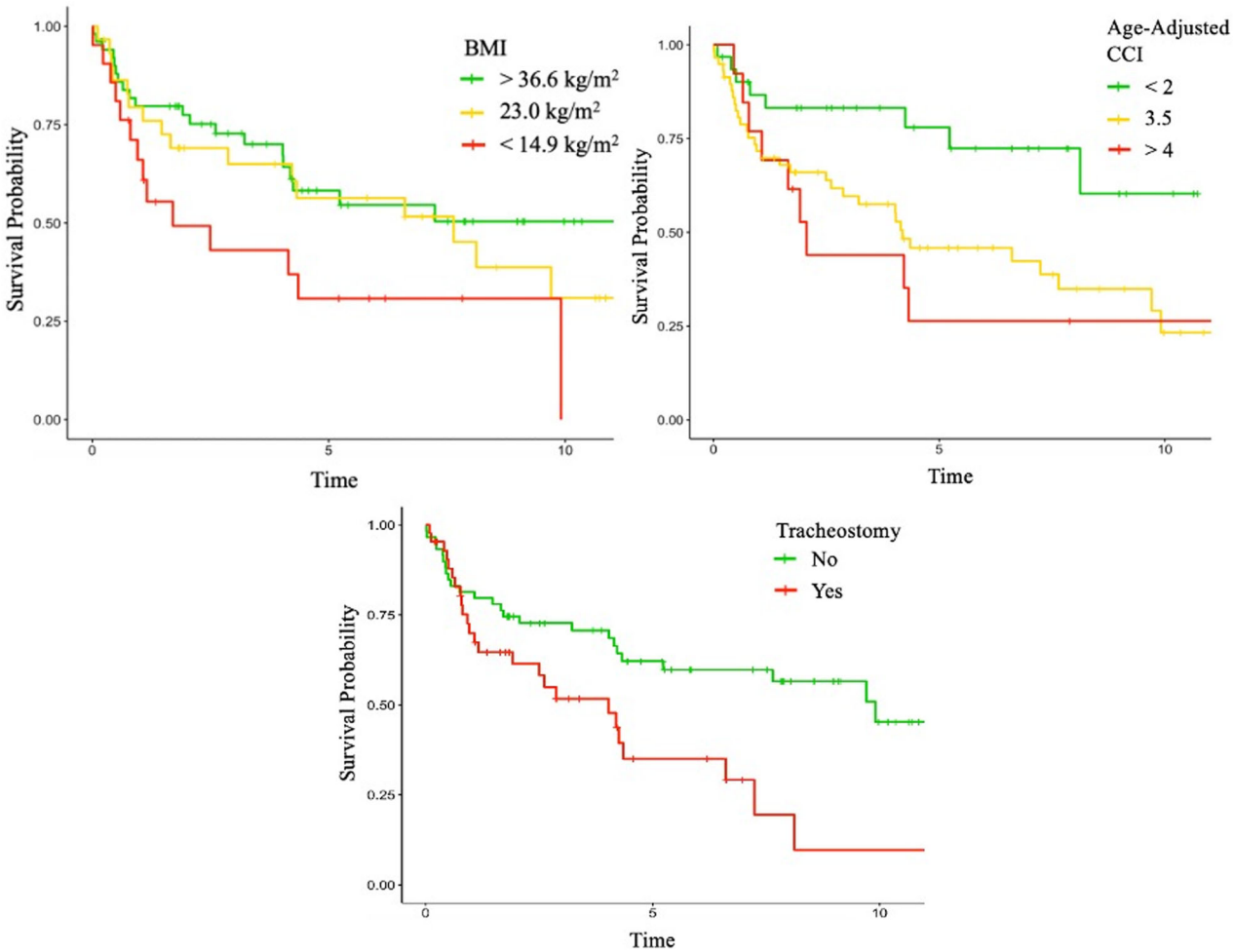



**Fig. 6** Kaplan–Meier curves of significant demographic and clinical features stratified by patient risk stratum

becomes more valuable than solely prioritizing predictive ability when making clinical decisions for individual patients.

This study has several limitations. First, the sample size ($n = 102$) was relatively small and originated from a single center, which may potentially limit the generalizability of the findings. However, it is worth noting that this cohort represents the largest group of LTx in SSc patients with CT scans to date. Given the scarcity of centers performing LTx in SSc patients and the low prevalence of SSc in LTx, finding an external cohort for validation presents a considerable challenge. To ensure the robustness of our findings, we performed 5-fold cross-validation over the cohort to minimize model overfitting. Additionally, the primary objective of our work was to introduce and validate CT-derived features as prognostic factors in SSc-related LTx. Second, there was missing clinical data. For instance, the data on primary graft dysfunction, the primary cause of early mortality, was incomplete and therefore not used in the study. Third, the CT scans were acquired over 16 years, resulting in significant variations in image quality due to the variations in imaging protocols. Our 3D-CNN body composition algorithms calculate tissue metrics over large areas of the body on CT and have been robustly validated against a large amount of CT scans acquired using different protocols [9, 25, 26], limiting the potential impact of data heterogeneity. However, we acknowledge the need for future testing. Another potential limitation was the large range of time from the CT scan date to the transplant date for patients of our cohort (Table 1). We acknowledge that the patient's body composition can change over time. The COVID-19 pandemic likely impacted the availability of CT imaging and scheduled surgeries in our cohort, resulting in delays in surgeries and pre-LTx assessments. Despite these limitations, our works successfully introduced a novel imaging modality to LTx survival prediction in cases of SSc.

## Conclusion

LTx in SSc patients is a complex procedure that requires accurate prediction of survival outcomes to guide appropriate pre-LTx screening procedures. Our study sheds light on the potential of utilizing CT-derived radiographic features to predict post-LTx survival in SSc patients before they undergo LTx. We discovered that bone density, muscle ratio, and artery–vein volume ratio are independent predictors of post-LTx survival. Furthermore, survival models incorporating these features demonstrated improved performance compared to traditional models based solely on patient demographics and clinical variables. The automated extraction of CT-derived features offers a more clinically feasible approach to manual annotations. Our robust methodologies can be applied to other time-to-event analyses, and the inclusion of CT-derived features offers valuable insight into whole-body risk factors not traditionally considered by clinicians.

### Abbreviations

| | |
|---|---|
| CI | Confidence interval |
| CPH | Cox proportional-hazards |
| HR | Hazard ratio |
| IMAT | Intramuscular adipose tissue |
| LTx | Lung transplant |
| ML | Machine learning |
| ROC-AUC | Area under the receiver operating characteristics curve |
| SAT | Subcutaneous adipose tissue |
| SM | Skeletal muscle |
| SSc | Systemic sclerosis |
| VAT | Visceral adipose tissue |

## Supplementary information

The online version contains supplementary material available at https://doi.org/10.1007/s00330-024-11077-9.

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## Compliance with ethical standards

### Guarantor

The scientific guarantor of this publication is Dr. Jiantao Pu.

### Conflict of interest

The authors of this manuscript declare no relationships with any companies whose products or services may be related to the subject matter of the article.

### Statistics and biometry

No complex statistical analyses were conducted.

### Informed consent

Written informed consent was waived by the Institutional Review Board.

### Ethical approval

Institutional Review Board approval was obtained from the University of Pittsburgh IRB #: 20100067.

### Study subjects or cohorts overlap

No study subjects or cohorts have been previously reported.

### Methodology

- Retrospective
- Observational
- Performed at one institution

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